

Miles Pines

Homework 2

9/15/25

Collaborators: None

Idea: CampusKey

CampusKey is a mobile and desktop app that connects to a student's LMS (Canvas, Blackboard, etc...), school email, and calendar to automatically ingest syllabi, assignments, and announcements. It turns these into an actionable plan with auto-generated study blocks, deadline reminders, and one-tap actions (e.g., "submit draft," "email TA," "book office hours"). Sensitive data is processed on-device by a small language model with optional encrypted cloud sync.

Similar services:

- **General planners:** Google Calendar/Tasks, Apple Reminders, Todoist, Notion are great at task capture but weak at parsing syllabi or LMS content automatically.
- **Student planners:** MyStudyLife, ClassUp focus on schedules but rely on manual entry; limited LMS deep integration.
- **LMS features:** Canvas/Blackboard calendars show due dates but do not create personalized study plans or cross-tool automations.
- **Agentic assistants:** AI chatbots (e.g., ChatGPT) help with Q&A but do not orchestrate tasks across school systems with strict privacy constraints.

Differentiation:

- On-device AI parsing of syllabi and announcements; privacy by default.
- Deep LMS + email + calendar connectors that *act* (draft emails, schedule office hours) rather than only display data.
- Planning engine turns workloads into adaptive time-boxed study blocks.

Why now:

- Mature LMS APIs and standardized calendar feeds enable integrations.
- Modern devices have NPUs/Neural Engines for small LLMs on-device.
- Passkeys/OAuth and scoped permissions make secure connections feasible.
- Rising expectations for privacy-preserving, explainable AI.

Stakeholders:

Stakeholder	Applies?	Values & Motivations	Expected Behaviors
Primary Users: Students	Yes	Less stress; privacy; fewer clicks; academic success.	Authorize connectors, review plans, give feedback.
Instructors/TAs	Sometimes	Fewer repetitive emails; students prepared.	Publish structured announcements, share office hours calendars.
Customers (Students)	Yes	Low price; real benefit; exportability.	Convert to paid; cancel if unclear; refer peers.
Universities/IT	Optional	Compliance; SSO; support.	Security review; pilot programs; bulk licensing.
Regulators	Yes	Privacy; data minimization; auditability.	Audit requests; mandate disclosures.
Owners/Founders	Yes	Product-market fit; growth.	Prioritize roadmap; raise capital.
Investors	Yes	Growth; unit economics.	Expect reporting; influence strategy.
Employees	Yes	Building a respected tool; clear scope.	Ship connectors; triage bugs.

Platform Partners	Yes	API usage compliance.	Provide/limit API scopes; co-marketing.
App Stores	Yes	User safety; subscription compliance.	Policy reviews; enforce rules.
Parents/Guardians	Sometimes	Student success; cost control.	Pay premium; request visibility.

User Journey: Primary User (Student)

Persona: Alex (19), first-year undergrad juggling four courses, a part-time job, and a club. Overwhelmed by scattered deadlines.

Stage	Alex's Goal & Actions	System Touchpoints	Values Tensions
Discover	Sees a friend's post; installs CampusKey.	App Store, marketing site.	Marketing wants contact data vs. student wants privacy.
Connect	Uses passkey; connects email, calendar, LMS via OAuth.	OAuth screens, permissions ledger.	Rich scopes vs. least privilege (regulators/university).
Ingest	CampusKey parses syllabi; Alex verifies course list.	On-device parser, previews.	Accuracy vs. speed/battery.
Plan	App proposes weekly study plan; Alex adjusts preferences.	Planner engine, calendar writes.	Automation vs. autonomy.
Act	Gets nudges with one-tap actions.	Notifications, deep links.	Helpful vs. notification fatigue.
Recover	Late shift triggers reflowed plan.	Rescheduling engine.	Optimal schedule vs. user preferences.
Finals	Focus mode with cumulative study plans.	Focus mode, analytics.	Reporting desired by others vs. privacy/consent.